AECOM

AI Solution Engineer Case

Technical Assessment & Deliverables Brief

This document outlines the case assignment provided to candidates for the AI Solution Engineer role at AECOM. It defines three deliverables and the live interview format, each designed to evaluate structured thinking, technical depth, and communication across stakeholder audiences.

Case Introduction

Client Overview

Southeastern Grid & Water (SGW) is a fictional US-based utility and infrastructure operator serving over **8 million residents** across coastal and inland regions vulnerable to hurricanes, flooding, heatwaves, and wildfires. SGW manages a large portfolio of critical infrastructure assets, including substations, transmission networks, water treatment facilities, and pumping stations.

In recent years, the company has faced rising operational costs, increased service disruptions, growing insurance premiums, and increasing regulatory pressure related to climate resilience and emergency preparedness.

The Challenge

AECOM has been engaged to help SGW modernize its operational resilience capabilities through an **AI-enabled decision support platform**. Today, operational data is fragmented across GIS systems, maintenance platforms, weather feeds, and field operations tools — making it difficult to:

- Proactively assess infrastructure risks
- Coordinate emergency response activities
- Provide leadership teams with real-time situational awareness during severe weather events

Candidate Guidance

Candidates should assume the client context is intentionally incomplete. The assessment is designed to evaluate how candidates:

- Identify missing information and fill gaps with explicit, reasoned assumptions
- Frame and prioritize problems under ambiguity
- Structure solutions that balance technical depth with business clarity

Clarity, prioritization, practical thinking, and communication are valued over polished production-ready deliverables.

Deliverable 1 — Product Requirements Document (PRD)

Prepare a concise but structured PRD intended for the **technical delivery team**. The PRD should demonstrate structured product thinking, technical clarity, prioritization, and the ability to communicate effectively with engineering teams.

Problem Definition & Business Context	Clearly articulate the operational problem and why it matters to SGW.
Key Assumptions & Unknowns	State all assumptions explicitly and explain how they shape the solution.
Target Users & Pain Points	Define the operational workflows and user groups most affected.
Functional & Non-Functional Requirements	Specify system behavior, performance, availability, and security needs.
Proposed AI Capabilities	Describe expected AI behaviors and how they improve decision-making.
High-Level Architecture & Integrations	Outline system components and integration with GIS, weather, and field ops tools.
Data Requirements & Dependencies	Identify required datasets, assumptions, and data quality considerations.
Security, Governance & Human-in-the-Loop	Address responsible AI usage, oversight mechanisms, and compliance.
Success Metrics, MVP Scope & Delivery Priorities	Define what success looks like and what ships first.

■ *Submit prior to the interview session.*

Deliverable 2 — Executive Management Briefing

Prepare a concise executive-level briefing for **senior leadership and decision-makers**. The audience is primarily concerned with business outcomes, implementation realism, governance, and strategic alignment — not low-level technical detail.

Strategic Business Value	Expected operational impact and case for AI-enabled resilience.
Financial Implications & ROI	Cost drivers, efficiency opportunities, and expected return.
Delivery Roadmap	Implementation phases, milestones, and major dependencies.
Governance & Compliance	Enterprise risk considerations and regulatory alignment.
Scalability	Expansion across regions, infrastructure domains, or additional workflows.

■ *Submit prior to the interview session.*

Deliverable 3 — Technical Prototype & Video Demo

Build and submit a **lightweight technical prototype** demonstrating at least one core workflow from the proposed solution. Mocked or simulated data is acceptable. The objective is to demonstrate practical solution engineering skills, rapid prototyping ability, and effective use of modern AI tooling.

1. Source Code Repository / Access

- All code required to run and assess the prototype
- README with setup and execution instructions
- Description of architecture, assumptions, and limitations
- Any sample or mocked data used in the demonstration

2. Video Demonstration (5–10 minutes)

- Demonstrate the working solution end-to-end
- Walk through at least one end-to-end user workflow
- Explain key technical decisions and architecture choices
- Highlight AI capabilities used within the solution
- Discuss assumptions, limitations, and future enhancement opportunities

Example Prototype Formats

- AI-assisted operational workflows
- Retrieval and reasoning workflows
- Interactive copilots or assistants
- Operational dashboards or alerting systems
- Risk scoring or prioritization systems
- Lightweight web applications or automation flows

Live Interview Session — Stakeholder Presentation & Working Demo

Candidates who successfully complete the deliverables will be invited to a live session presenting to a mixed audience representing **operational stakeholders, technical teams, and leadership.**

Presentation Expectations

- Clearly explain the operational and business challenges
- Demonstrate the proposed solution and future-state workflows
- Show how the solution improves resilience, preparedness, and emergency response
- Include visuals, workflows, architecture diagrams, mockups, or prototypes
- Address operational adoption, usability, and organizational considerations
- Demonstrate practical solution engineering skills and rapid prototyping ability

AI Capabilities of Interest

Candidates are encouraged to identify opportunities for AI capabilities **beyond LLMs**, including:

- Forecasting
- Anomaly Detection
- Optimization
- Computer Vision
- Predictive ML Models
- Reinforcement Learning

Session Format

Phase	Duration	Focus
Presentation & Demo	~30 min	Walkthrough, workflow demo
Discussion & Q&A;	~30 min	Trade-offs, tooling, technical Q&A;

The goal is to evaluate communication skills, stakeholder engagement, rapid prototyping ability, technical reasoning, and the ability to discuss trade-offs in a collaborative setting.